

Course Syllabus

In this course you will learn:

Module 1: Introduction to JavaScript and Computer Programming

- About JavaScript (how to communicate with the computer, what is JS, advantages and limitations of JS, where is JS used today)
- Setting up the programming environment (development tools, online development environment, local development environment)
- First JS program – Hello, World! (a few words about HTML, how to run your JavaScript code, executing the code directly in the console)

Module 2: Variables, Data Types, Type Casting, and Comments

- Variables (naming, declaring and initializing variables, declarations and strict mode, changing variable values, constants, scope)
- Primitive data types (Boolean, Number, BigInt, String, undefined, null, type casting – primitive construction functions and primitive conversions, implicit conversions)
- Complex data types (Object, Array, basic Array properties and methods)
- Comments (single-line comments, multi-line comments, documentation)

Module 3: Operators and User Interaction

- Assignment, arithmetic, and logical operators (what are operators, assignment operators, arithmetic operators, logical operators, compound assignment operators)
- Strings, comparison, and other JS operators (string concatenation and compound assignments, comparison operators, conditional operators, typeof, instanceof and delete operators, operator precedence)
- Interacting with the user (dialog boxes – alert, confirm, prompt)

Module 4: Control Flow – Conditional Execution and Loops

- Conditional execution (what is conditional execution, the if–else statement, the conditional operator, the switch–case statement)
- Loops (what are loops, the while loop, the do–while loop, the for loop, the for–of loop, the for–in loop, the break and continue statements)

Module 5: Functions

- Function basics (what are functions, declaring functions, calling functions, local variables, the return statement, function parameters, shadowing)
- Functions as first-class members (function expressions, passing a function as a parameter, callbacks)
- Arrow functions (declaring and calling)
- Recursion (basic idea)

Module 6: Errors, exceptions, debugging, and troubleshooting

- Errors and exceptions – introduction (natural languages and communication errors, errors vs. exceptions, errors without exceptions, limited confidence)
- Basic types of errors in JS (SyntaxError, ReferenceError, TypeError, RangeError)
- Exception handling (the try–catch statement, the finally statement, the throw statement, and custom errors)
- Code debugging and troubleshooting (what is debugging, step-by-step execution, viewing and modifying variables, the step out option, measuring code execution time)